

MIHIR MAKWANA

Computer Vision · Deep Learning · AI

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PROFILE

MSc Artificial Intelligence student at Kingston University with hands-on computer vision experience and a year of full-stack development in production environments. Skilled in PyTorch, FaceNet, MTCNN, and OpenCV through a group Smart Office Face Recognition System project, and in MERN/Laravel stacks through prior internship work. Seeking AI/ML internships, placements, and part-time roles in London — with the engineering background to take models from notebook to deployed product.

EDUCATION

MSc Artificial Intelligence, Kingston University, London Jan 2026 – Jan 2028
Artificial Intelligence with Professional Placement

- **Relevant modules:** Computer Vision (CI7523), Deep Learning, Machine Learning, Natural Language Processing, AI Ethics
- **Key project:** Smart Office Face Recognition System — PyTorch, FaceNet, MTCNN, RetinaFace

Master of Computer Applications (MCA), ISTAR College, Gujarat Jul 2023 – May 2025
CGPA: 9.27 / 10 · Class Representative, MCA 2nd Year · Coursework: React, Node.js, Laravel, MongoDB, Python

Bachelor of Computer Applications (BCA), SEMCOM College, Gujarat Jun 2020 – Apr 2023
CGPA: 8.67 / 10 · Coursework: JavaScript, Java, PHP, Python, AI & Machine Learning

TECHNICAL SKILLS

AI / ML	PyTorch, TensorFlow / Keras, scikit-learn, OpenCV, FaceNet, MTCNN, Streamlit, Deep Learning, Computer Vision, CNNs, Transfer Learning (MobileNetV2, ResNet50), Feature Engineering, Imbalanced Data (SMOTE)
Programming	Python, JavaScript, Java, PHP, C++, SQL
Web / Full-Stack	React.js, Node.js, Express.js, Laravel, MongoDB, MySQL, REST APIs, HTML5, CSS
Tools	Git, GitHub, VS Code, Jupyter, Postman, Agile / Scrum

PROJECTS

Smart Office Face Recognition System · Kingston University (CI7523) Jan – Apr 2026
Role: Face Detection Specialist · Team of 4

- Built a real-time face recognition system for office access control using MTCNN (facenet-pytorch) and RetinaFace for detection, FaceNet and ArcFace for recognition.
- Implemented preprocessing pipeline with CLAHE and Gaussian denoising to improve robustness under varied lighting conditions.
- Migrated detection backbone from TensorFlow MTCNN to facenet-pytorch, resolving numpy dtype and Windows Unicode encoding issues.
- Standardised embedding database to consistent 512D FaceNet vectors, eliminating silent match failures from mixed dimensions.
- Optimised Streamlit demo performance using `@st.cache_resource`, reducing model reload on every rerun.

Stack: Python · PyTorch · facenet-pytorch · OpenCV · Streamlit
github.com/mihirmakwana03/CV-Assignment

Multi-Class Classification on Imbalanced Dataset · Kingston University (CI7521 Machine Learning & Deep Learning — CW1) Feb – Mar 2026

- Built and compared 8 classifiers (LDA, QDA, Decision Tree, KNN, Logistic Regression, SVM, Random Forest, Gaussian NB) on a 9.8K-sample, 32-feature, 5-class imbalanced dataset (OpenML #4538).
- Tuned hyperparameters with RandomizedSearchCV → GridSearchCV; compared scalers (Quantile, Power, Standard) and resampling strategies (SMOTE oversampling / undersampling) to handle class imbalance.
- Evaluated using balanced accuracy, macro/micro one-vs-rest ROC-AUC, and confusion matrices; used LinearSVC-based feature selection to improve model generalisation.

Stack: Python · scikit-learn · imbalanced-learn · pandas · matplotlib

Medical Image Classification — OrganSMNIST+ · Kingston University (CI7521 Machine Learning & Deep Learning — CW2) Mar 2026

- Classified abdominal CT organ images from MedMNIST+ OrganSMNIST at 224×224 resolution, benchmarking MLPs, CNNs, and transfer-learning models across 13 experiments in a 4-person team.
- Trained a baseline MLP plus 5 improved variants (depth/width, dropout, batch norm, SGD+momentum, L2/AdamW); built baseline and deeper CNNs with matching ablations.
- Applied transfer learning with MobileNetV2 and ResNet50 (ImageNet weights); added a data-augmentation variant that reduced overfitting on the best CNN configuration.
- Evaluated 6 final models on the held-out test set using accuracy, F1, precision, recall, and confusion matrices — ResNet50 gave the strongest generalisation.

Stack: Python · TensorFlow / Keras · MedMNIST · MobileNetV2 · ResNet50 · NumPy · Matplotlib

- Built React.js frontend for a MERN-stack CRM tracking solar-system clients and service tickets; consumed REST APIs from Node.js/Express backend with MongoDB.
 - Designed reusable dashboard components, form validation, and role-based views for admin and field-engineer users.
- Stack:** React.js · Node.js · Express.js · MongoDB
github.com/mihirmakwana03/growatt-frontend · growatt-frontend.vercel.app

- Built and maintained responsive Laravel website with Blade templates and Tailwind-styled frontend; integrated Groove.cm for headless content management.
 - Optimised cross-platform performance and collaborated with stakeholders on brand-aligned UX.
- Stack:** Laravel · PHP · Blade · JavaScript · Tailwind CSS
github.com/mihirmakwana03/HF-Testing · happinessfactors.laravel.cloud

EXPERIENCE

Delivered the company's Laravel site end-to-end (see Projects) and edited promotional video content to support digital marketing.

Shipped a MERN-stack CRM for Growatt InfoSystem in an Agile team (see Projects) — React frontend, Express APIs, MongoDB schemas, daily stand-ups and code reviews.

Certifications: Introduction to AI · Exploratory Data Analysis for ML · Complete Git & GitHub — *Coursera, Geekster*
Achievements: 1st Place, Poster Presentation — IGNITE, CHARUSAT (Jan 2024) · Technical Event Coordinator, TECHNOKHOJ-2024
Languages: English (Professional) · Hindi (Native) · Gujarati (Native)